

Skills checks cover pre-requisite material needed for the second unit of the course, derivatives. Answer the following questions neatly and completely. Calculators are not allowed.

1. Express the function $y = \left(\sqrt[3]{x+6}\right)^2 + 2$ as a composition of functions $f(g(x))$.

Many answers are possible.

$$f(x) = x^2 + 2 \quad \text{and} \quad g(x) = \sqrt[3]{x+6}$$

2. Express the function $y = \frac{1}{2} \cos(x^3 + 2)$ as a composition of functions $f(g(x))$.

Many answers are possible.

$$f(x) = \frac{1}{2} \cos(x) \quad \text{and} \quad g(x) = x^3 + 2$$

3. Express the function $y = \sqrt{e^x - 2}$ as a composition of functions $f(g(x))$.

Many answers are possible.

$$f(x) = \sqrt{x} \quad \text{and} \quad g(x) = e^x - 2$$

4. Solve the following equation for y : $4x^2 - 3x + xy = 2y + 5x^2y$.

$$4x^2 - 3x + xy = 2y + 5x^2y$$

$$xy - 2y - 5x^2y = 3x - 4x^2$$

$$y(x - 2 - 5x^2) = 3x - 4x^2$$

$$y = \frac{3x - 4x^2}{x - 2 - 5x^2}$$

5. Solve the following equation for f : $\frac{1}{f} + \frac{1}{g} = \frac{1}{h}$.

$$\frac{1}{f} + \frac{1}{g} = \frac{1}{h} \rightarrow \frac{1}{f} = \frac{1}{h} - \frac{1}{g} \rightarrow 1 = \left(\frac{1}{h} - \frac{1}{g}\right)f \rightarrow f = \frac{1}{\left(\frac{1}{h} - \frac{1}{g}\right)}$$

6. Solve the following equation for y : $\cos(x)(y+1) = 5y-3$.

$$\cos(x)(y+1) = 5y-3$$

$$\cos(x)y + \cos(x) = 5y-3$$

$$\cos(x)y - 5y = -3 - \cos(x)$$

$$y[\cos(x)-5] = -3 - \cos(x)$$

$$y = \frac{-3 - \cos(x)}{\cos(x)-5}$$

7. Find the equation of the line that contains the point $(2,4)$ and is parallel to the line $2y = 7x - 4$.

$$\text{We see that } 2y = 7x - 4 \rightarrow y = \frac{7}{2}x - 2.$$

Therefore, our line must have

$$\text{Point: } (2,4)$$

$$\text{Slope: } \frac{7}{2}$$

$$\text{Equation: } y - 4 = \frac{7}{2}(x - 2) \rightarrow y = \frac{7}{2}x + 1$$

8. A function $f(x)$ is linear and passes through the point $(-3,1)$. If the slope of the line is 2, find the value of $f(6)$.

Our line must have

$$\text{Point: } (-3,1)$$

$$\text{Slope: } 2$$

$$\text{Equation: } y - 1 = 2(x + 3) \rightarrow y = 2x + 7 \rightarrow f(x) = 2x + 7$$

$$\text{Therefore, } f(6) = 2(6) + 7 = 19$$

9. Find the equation of the linear function that contains the point $(2,3)$ and is perpendicular to the line $y + 7 = 3x$.

$$\text{We see that } y + 7 = 3x \rightarrow y = 3x - 7.$$

Therefore, our line must have

$$\text{Point: } (2,3)$$

$$\text{Slope: } -\frac{1}{3}$$

$$\text{Equation: } y - 3 = -\frac{1}{3}(x - 2) \rightarrow y = -\frac{1}{3}x + \frac{11}{3}$$

10. Expand the following using properties of logarithms: $\ln\left(\frac{\sqrt{x}}{x^5(x-2)}\right)$.

$$\ln\left(\frac{\sqrt{x}}{x^5(x-2)}\right) = \ln(\sqrt{x}) - \ln(x^5(x-2)) = \frac{1}{2}\ln(x) - [\ln(x^5) + \ln(x-2)] = \frac{1}{2}\ln(x) - [5\ln(x) + \ln(x-2)]$$

11. Expand the following using properties of logarithms: $\ln\left(\sqrt[3]{\frac{x+3}{x-5}}\right)$.

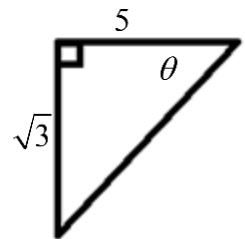
$$\ln\left(\sqrt[3]{\frac{x+3}{x-5}}\right) = \frac{1}{3}\ln\left(\frac{x+3}{x-5}\right) = \frac{1}{3}[\ln(x+3) - \ln(x-5)]$$

12. Expand the following using properties of logarithms: $\ln\left(\sqrt{x^2(x+1)(x+2)}\right)$.

$$\begin{aligned}\ln\left(\sqrt{x^2(x+1)(x+2)}\right) &= \frac{1}{2}\ln(x^2(x+1)(x+2)) = \frac{1}{2}[\ln(x^2) + \ln(x+1) + \ln(x+2)] \\ &= \frac{1}{2}[2\ln(x) + \ln(x+1) + \ln(x+2)]\end{aligned}$$

13. Create a trig. ratio that corresponds to the triangle.

$$\tan(\theta) = \frac{\sqrt{3}}{5}$$



14. Compute the exact value of $\tan\left(\frac{4\pi}{3}\right)$.

$$\tan\left(\frac{4\pi}{3}\right) = \frac{\left(-\frac{\sqrt{3}}{2}\right)}{\left(-\frac{1}{2}\right)} = \sqrt{3}$$

15. Create a trig. ratio that corresponds to the triangle.

$$\sin(\theta) = \frac{\pi}{\sqrt{2}}$$

